EDC mock B

Answers and worked solutions · Day 28 · Fri 20 Nov · pack EX2

Marking guide

Mark strictly. Give a mark only when your script shows the line that earns it. Fill in the table at the end, then use the “reread” column to plan the morning.

Question 1 (8 marks)

Part	Answer	Why
I	**b**	$p = n_i^2/N_D = (1.5\times10^{10})^2/10^{16} = 2.25\times10^4$ cm^{-3}
II	**c**	Bridge: PIV $= V_{s,\text{peak}} - V_D \approx V_m$. Centre tap: PIV $= 2V_s - V_D \approx 2V_m$
III	**b**	Saturation: both junctions forward biased, the collector cannot take βI_B, so the forced β is smaller
IV	**c**	$I_D = 6(1-1/(-3))^2 = 6(4/3)^2 = 10.7$ mA > I_{DSS}. The gate is insulated, so no gate current flows
V	**b**	$NM_L = V_{IL} - V_{OL} = 2.1 - 0.2 = 1.9$ V; $NM_H = V_{OH} - V_{IH} = 4.8 - 2.9 = 1.9$ V
VI	**c**	$20\log_{10}(4000/0.4) = 20\log_{10}10^4 = 80$ dB
VII	**b**	Each half needs $V_{BE} \approx 0.7$ V before it conducts, so the output is flat near zero crossing
VIII	**b**	$1/(1+2/50) = 0.962$

> **COMMON MISTAKES:** in V, swapping V_{IL} and V_{IH} (option c is $V_{IH} - V_{IL}$, the width of the transition region). In II, quoting the centre-tap PIV for the bridge.

REREAD: I E1; II E3; III E5; IV E9/E10; V E9; VI and VII E13; VIII E8.

Question 2 (7 marks)

(a)(i) p-type with $N_A \gg n_i$: $p \approx N_A = 5 \times 10^{15}$ cm^{-3}, and

$$n = \frac{n_i^2}{p} = \frac{(1.5 \times 10^{10})^2}{5 \times 10^{15}} = \mathbf{4.5 \times 10^{4}\ cm^{-3}} \quad [1]$$

(a)(ii) The electron term is negligible:

$\sigma = q(n\mu_n + p\mu_p) \approx 1.6 \times 10^{-19} \times 5 \times 10^{15} \times 500 = \mathbf{0.400\ (\Omega\,cm)^{-1}}$, $\quad \rho = 2.50\ \Omega\,\text{cm} \quad [1]$

Work in cm: $L = 0.2$ cm, $A = 0.01 \times 0.01 = 10^{-4}$ cm².

$$R = \frac{\rho L}{A} = \frac{2.5 \times 0.2}{10^{-4}} = \mathbf{5.00\ k\Omega} \quad [1]$$

(a)(iii) $T = 200 + 273.15 = 473.15$ K. Use k in eV/K in the exponent:

$$n_i = 7.3 \times 10^{15}(473.15)^{3/2} e^{-1.12/(2\times 8.62\times 10^{-5}\times 473.15)} = \mathbf{8.18 \times 10^{13}\ cm^{-3}}$$

This is 1.6 % of N_A, so the bar is **still extrinsic**: $p \approx 5.00 \times 10^{15}$ cm^{-3} and $n = n_i^2/p \approx 1.34 \times 10^{12}$ cm^{-3}. [1]

(Check: the same formula at 300 K gives 1.51×10^{10} cm^{-3}, which matches the given room-temperature value.)

(b)(i) At 27 °C, $V_T = \dfrac{1.38 \times 10^{-23} \times 300}{1.6 \times 10^{-19}} = 25.875$ mV.

$$I_D = \frac{5 - 0.7}{4.3\ \text{k}} = \mathbf{1.000\ mA}, \qquad r_d = \frac{nV_T}{I_D} = \mathbf{25.9\ \Omega} \quad [1]$$

For the ac signal, the 5 V supply is a short and the diode is r_d. That gives a divider:

$$v_d = 20\ \text{mV} \times \frac{25.9}{4300 + 25.9} = \mathbf{0.120\ mV\ peak} \quad [1]$$

(b)(ii) $\Delta T = 50$ °C, so $V_D = 0.7 - 0.0025 \times 50 = 0.575$ V.

$I_D = (5 - 0.575)/4.3\ \text{k} = 1.029$ mA. $V_T = 1.38 \times 10^{-23} \times 350/1.6 \times 10^{-19} = 30.19$ mV. So $r_d = 30.19/1.029 = \mathbf{29.3\ \Omega}$ and $v_d = 20 \times 29.3/4329 = \mathbf{0.136\ mV}$ peak.

r_d rises because $V_T \propto T$ rises by 17 %, while I_D rises by only 3 % (the 4.3 kΩ resistor nearly fixes the current). [1]

COMMON MISTAKES: using $k = 1.38 \times 10^{-23}$ with E_g in eV in the exponent; forgetting °C → K; using L in mm and A in cm²; leaving out the 4.3 kΩ divider for v_d (the answer is not 20 mV).

REREAD: E1 (carrier concentrations, $n_i(T)$), E2 (small-signal r_d, temperature rules).

Question 3 (5 marks)

(a) Find each diode's turn-on condition from the node voltage.

- D1 (anode at v_o, cathode at +4 V) conducts when $v_o \geq 4 + 0.7 =$ **+4.7 V**. Then v_o is held at +4.7 V. [1]
- D2 (cathode at v_o, anode at −6 V) conducts when $v_o \leq -6 - 0.7 =$ **−6.7 V**. Then v_o is held at −6.7 V. [1]
- In between, both diodes are off, no current flows in the 1 kΩ resistor, and $v_o = v_i$.

Transition angles from $12 \sin \omega t$:

- Positive clipping: $\sin^{-1}(4.7/12) = 23.1^\circ$. Clipped from **23.1° to 156.9°**.
- Negative clipping: $\sin^{-1}(6.7/12) = 33.9^\circ$. Clipped from $180^\circ + 33.9^\circ =$ **213.9° to 326.1°**.

SKETCH DESCRIPTION: the output follows the sine from 0 V up to +4.7 V at 23.1°. It stays flat at +4.7 V until 156.9°, then follows the sine down through 0 V at 180° to −6.7 V at 213.9°. It stays flat at −6.7 V until 326.1°, then follows the sine back up to 0 V at 360°. The output is an asymmetric flat-topped sine. Its top is at +4.7 V, its bottom is at −6.7 V, and it is 11.4 V peak-to-peak. [1]

(b) Start in the half-cycle where the diode conducts. In the +8 V half, the output tries to rise, so the diode turns on and holds $v_o = 2 + 0.7 =$ **+2.7 V**. The capacitor charges to $8 - 2.7 = 5.3$ V (input side positive).

In the −8 V half, the diode turns off, and the capacitor's 5.3 V adds to the input: $v_o = -8 - 5.3 =$ **−13.3 V**. The diode is reverse biased (−13.3 V < 2.7 V), which confirms it is off. [1]

SKETCH DESCRIPTION: a square wave of the same shape and frequency (1 kHz). It sits at +2.7 V for the first half-period and at −13.3 V for the second. It is **16 V peak-to-peak**, the same as the input, and its average is −5.3 V. The whole input waveform has been shifted down by 5.3 V.

$RC = 100\ \text{k} \times 1\ \mu\text{F} = 100$ ms, which is 200 times the 0.5 ms half-period. The capacitor barely discharges, so the output is flat (tilt under 0.5 %). [1]

> **COMMON MISTAKES:** placing the clip at the battery voltage without the 0.7 V; getting the sign of the D2 level wrong (read which battery terminal the diode touches); for the clamper, starting the analysis in the half-cycle where the diode is off; giving an output peak-to-peak value different from the input's.

REREAD: E3 (clippers and clampers, playbooks E and F).

Question 4 (5 marks)

1. Secondary peak $V_{s,p} = 12\sqrt{2} = 16.97$ V. Two diodes conduct in series in a bridge: $V_P = 16.97 - 1.4 =$ **15.57 V**. [1]
2. Full-wave, so use $2f$: $C = \dfrac{V_P}{2fRV_r} = \dfrac{15.57}{2 \times 50 \times 150 \times 0.8} =$ **1297 µF**. [1]
3. With 1500 µF: $V_r = \dfrac{15.57}{2 \times 50 \times 150 \times 1500 \times 10^{-6}} =$ **0.692 V** p-p. $V_{dc} = 15.57 - 0.346 =$ **15.22 V**. [1]
4. $I_L \approx V_P/R = 103.8$ mA, and $\sqrt{V_P/(2V_r)} = \sqrt{15.57/1.384} = 3.354$. So $i_{Dmax} = 103.8(1 + 2\pi \times 3.354) =$ **2.29 A**. (The average during conduction is $i_{Dav} = 103.8(1 + \pi \times 3.354) = 1.20$ A.) The conduction angle is $\omega\Delta t = \sqrt{2 \times 0.692/15.57} = 0.298$ rad = 17.1°. [1]
5. In a bridge, the reverse-biased diode sees the secondary peak minus one conducting drop: PIV $= 16.97 - 0.7 =$ **16.3 V**. [1]

If you used $I_L = V_{dc}/R = 101.5$ mA in part 4, you get $i_{Dmax} = 2.24$ A. Give the mark if the method is stated.

> **COMMON MISTAKES:** using f instead of $2f$ (that doubles C); subtracting only one 0.7 V for a bridge; using the half-wave i_{Dmax} form; quoting the centre-tap PIV ($2V_m$).

REREAD: E3 (filter design, playbook D).

Question 5 (8 marks)

(a) Thevenin at the base:

$$V_{Th} = 12 \times \frac{47}{15+47} = 9.097\ \text{V}, \qquad R_{Th} = 15\ \text{k} \parallel 47\ \text{k} = 11.37\ \text{k}\Omega \quad [1]$$

pnp loop from the +12 V rail, through R_E, across E–B, through R_{Th} to V_{Th}. Base current flows **out** of the base:

$$12 - (\beta+1)I_B R_E - 0.7 - I_B R_{Th} = V_{Th}$$

$$I_B = \frac{12 - 0.7 - 9.097}{11.37\ \text{k} + 101 \times 1.5\ \text{k}} = \frac{2.203}{162.9\ \text{k}} = 13.53\ \mu\text{A} \quad [1]$$

$I_C = 100 I_B =$ **1.353 mA**, $I_E = 1.366$ mA. [1]

$V_E = 12 - 1.366 \times 1.5 = 9.95$ V. $V_C = I_C R_C = 1.353 \times 2.7 =$ **3.65 V**.
$V_{EC} = 9.95 - 3.65 =$ **6.30 V**. Since 6.30 V > 0.2 V, the transistor is **active**. [1]

(b)

β	I_C (mA)	Change from β = 100
50	1.254	−7.3 %
100	1.353	—
200	1.408	+4.1 %

A fourfold spread in β moves I_C by only about 11 % in total, so the bias is **stable**. The reason is that $(\beta+1)R_E = 151.5$ kΩ is about 13 times R_{Th} at β = 100. Even at β = 50 it is still about 6.7 times R_{Th}, so $I_C \approx (V_{CC} - V_{Th} - 0.7)/R_E$ almost independently of β. [2]

(c) Assume active: $V_C = 1.353 \times 8.2 = 11.09$ V, which is **above** $V_E = 9.95$ V. That would make $V_{EC} = -1.14$ V, which is impossible, so the transistor is **saturated**. [1]

Saturated model: $V_{EC} = 0.2$ V and $I_C \neq \beta I_B$. Two unknowns, I_B and I_C:

- Base loop: $12 - (I_B + I_C)(1.5\ \text{k}) - 0.7 = 9.097 + I_B(11.37\ \text{k})$
- Collector loop: $12 - (I_B + I_C)(1.5\ \text{k}) - 0.2 = I_C(8.2\ \text{k})$

Solving: $I_B = 29.9\ \mu$A and $I_C =$ **1.21 mA**. Check: $I_C/I_B = 40.5 < \beta = 100$, so it is saturated. $V_E = 10.14$ V and $V_C = 9.94$ V. [1]

(The largest R_C for active operation at β = 100 is $(9.95 - 0.2)/1.353\ \text{mA} = 7.21$ kΩ.)

COMMON MISTAKES: npn sign habits on a pnp (base current direction, V_{EB} not V_{BE}); using V_{CE} and getting a negative "active" answer without spotting it; in (c), keeping $I_C = \beta I_B$ after deciding the transistor is saturated.

REREAD: E4 (Thevenin two-loop method), E5 (pnp, β sensitivity, guess–analyse–check).

Question 6 (8 marks)

(a) DC:

$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta R_C} = \frac{10 - 0.7}{180\ \text{k} + 150 \times 2.2\ \text{k}} = 18.24\ \mu\text{A}, \quad I_E = 151 I_B = \mathbf{2.754\ mA} \quad [1]$$

$r_e = 26\ \text{mV}/2.754\ \text{mA} = \mathbf{9.44\ \Omega}$. ($V_{CE} = 10 - 2.754 \times 2.2 = 3.94$ V, so it is active.) [1]

AC:

$$Z_o \approx R_C \parallel R_F = 2.2\ \text{k} \parallel 180\ \text{k} = \mathbf{2.17\ k\Omega}$$

$$A_v = -\frac{R_C \parallel R_F}{r_e} = -\frac{2173}{9.44} = \mathbf{-230} \quad [1]$$

$$Z_i = \frac{r_e}{\frac{1}{\beta} + \frac{R_C}{R_C + R_F}} = \frac{9.44}{0.00667 + 0.01207} = \mathbf{504\ \Omega} \quad [1]$$

Miller check: $\beta r_e \parallel R_F/(1 - A_v) = 1416 \parallel 779 = 502\ \Omega$. An exact nodal solution of the r_e model gives $A_v = -230.2$ and $Z_i = 502\ \Omega$.

(b) DC: the base is at 0 V, so $V_E = -0.7$ V and

$$I_E = \frac{-0.7 - (-4)}{1.2\ \text{k}} = \mathbf{2.75\ mA}, \qquad r_e = \frac{26\ \text{mV}}{2.75\ \text{mA}} = \mathbf{9.45\ \Omega} \quad [1]$$

($V_C = 12 - 0.99 \times 2.75 \times 3.3 = 3.02$ V, so $V_{CB} > 0$ and it is active.)

$Z_i = R_E \parallel r_e = 1200 \parallel 9.45 = \mathbf{9.38\ \Omega}$. [1]

$$A_{v_L} = \frac{\alpha(R_C \parallel R_L)}{r_e} = \frac{0.99 \times (3.3\ \text{k} \parallel 6.8\ \text{k})}{9.45} = \mathbf{232.6} \quad [1]$$

$$A_{v_s} = \frac{Z_i}{Z_i + R_s} A_{v_L} = \frac{9.38}{59.38} \times 232.6 = \mathbf{36.8}$$

v_o is **in phase** with v_s (CB does not invert). The 50 Ω source throws away 84 % of the signal, because Z_i is only 9.4 Ω. [1]

> **COMMON MISTAKES:** in (a), using $R_F + \beta(R_C + R_E)$ with an R_E that does not exist, or using $A_v = -R_C/r_e$ without $\parallel R_F$ (this gives −233, which is close enough to score only if the formula is shown); in (b), putting a minus sign on the CB gain, or forgetting the source divider.

REREAD: E6 (r_e model, DC→AC), E7 (sections 1.5 and 1.6).

Question 7 (5 marks)

1. KCL at Q1's collector: I_{REF} supplies I_{C1} plus both base currents. With matched transistors, $I_{C1} = I_{C2} = I_o$, so $I_{REF} = I_o(1 + 2/\beta) = 2.00 \times 1.02 = 2.04$ mA. [1] Q1's collector is at $V_{B1} = -9 + 0.7 = -8.3$ V, so $R = \dfrac{0 - (-8.3)}{2.04\ \text{mA}} =$ **4.07 kΩ**. [1]
2. Q2 stays active while $V_{C2} \geq V_{E2} + 0.2 = -8.8$ V. Since $V_{C2} = 0 - I_o R_L$, the condition is $R_L \leq 8.8/2\ \text{mA} =$ **4.40 kΩ**. [2]
3. I_{REF} is set by R alone: still 2.04 mA. Then $I_o = 2.04/(1 + 2/50) =$ **1.96 mA**. Halving β changes I_o by only 1.9 %. [1]

> **COMMON MISTAKES:** using $1/\beta$ instead of $2/\beta$; measuring R's voltage from the −9 V rail instead of from Q1's collector (−8.3 V); forgetting that the compliance limit is set by $V_{CE,\text{sat}}$, not by V_{BE}.

REREAD: E8 (playbook F, current mirror).

Question 8 (8 marks)

(a) No gate current flows, so $V_G = 18 \times \dfrac{20}{120} = \mathbf{3.00\ V}$. The bias line is $V_{GS} = V_G - I_D R_S = 3 - 300 I_D$. [1]

Substitute into Shockley:

$$I_D = 8\ \text{mA}\left(1 - \frac{3 - 300 I_D}{-4}\right)^2 = 8\ \text{mA}\left(\frac{7 - 300 I_D}{4}\right)^2$$

This gives the quadratic $45\, I_D^2 - 3.1\, I_D + 0.0245 = 0$ (with I_D in A), with roots $I_D = 59.8$ mA and **9.11 mA**. [1]

Reject 59.8 mA: it gives $V_{GS} = 3 - 17.9 = -14.9$ V, which is beyond $V_P = -4$ V (cut off), so it is not a valid operating point. Keep

$$I_{DQ} = \mathbf{9.11\ mA}, \qquad V_{GSQ} = 3 - 0.300 \times 9.11 = \mathbf{+0.268\ V} \quad [1]$$

V_{GS} is positive and $I_D > I_{DSS}$: the device is in enhancement mode.

(b) $V_S = 9.11 \times 0.3 = 2.73$ V. $V_D = 18 - 9.11 \times 1.0 = 8.89$ V. $V_{DS} = 6.16$ V. Saturation requires $V_{DS} \geq V_{GS} - V_P = 0.268 + 4 = 4.27$ V, and 6.16 V meets this. [1]

(c) $g_{m0} = 2(8)/4 = \mathbf{4.00\ mS}$. $g_m = 4\left(1 - \dfrac{0.268}{-4}\right) = \mathbf{4.27\ mS}$. $g_m > g_{m0}$ because the Q-point is above I_{DSS} on the transfer curve, where the curve is steeper than at $V_{GS} = 0$. [1]

(d) $r_d = 1/10\ \mu\text{S} = 100$ kΩ.

- $Z_i = R_1 \parallel R_2 = 100\ \text{M} \parallel 20\ \text{M} = \mathbf{16.7\ M\Omega}$ [1]
- $Z_o = r_d \parallel R_D = 100\ \text{k} \parallel 1\ \text{k} = \mathbf{990\ \Omega}$ [1]
- $A_v = -g_m(r_d \parallel R_D) = -4.27\ \text{mS} \times 990 = \mathbf{-4.23}$; with $R_L = 3.3$ kΩ: $A_{v_L} = -4.27\ \text{mS} \times (100\ \text{k} \parallel 1\ \text{k} \parallel 3.3\ \text{k}) = \mathbf{-3.25}$ [1]

> **COMMON MISTAKES:** taking the larger root because it "looks like more current"; dropping the minus on V_P when it is substituted ($1 - V_{GS}/V_P$ with $V_P = -4$ becomes $1 + V_{GS}/4$); assuming V_{GS} must be negative and forcing a sign; including R_S in the AC gain although it is bypassed.

REREAD: E10 (D-MOSFET bias, quadratic method), E11 (g_m, CS amplifier).

Question 9 (12 marks)

(a)(i) Self-bias line: $V_{GS} = -I_D R_S = -2000 I_D$. Shockley:

$$I_D = 10\ \text{mA}\left(1 - \frac{2000 I_D}{4}\right)^2$$

The roots are 3.12 mA and **1.28 mA**. 3.12 mA gives $V_{GS} = -6.23$ V, which is beyond V_P, so reject it. [1]

$I_{DQ} =$ **1.283 mA** and $V_{GSQ} =$ **−2.57 V**. ($V_{DS} = 15 - 2.57 = 12.4$ V, so it is saturated.) [1]

(ii) $g_{m0} = 2(10)/4 = 5$ mS. $g_m = 5\left(1 - \dfrac{-2.567}{-4}\right) =$ **1.79 mS**. [1]

(iii) $Z_i = R_G =$ **1 MΩ**. $r_d = 50$ kΩ. $Z_o = r_d \parallel R_S \parallel \dfrac{1}{g_m} = 50\ \text{k} \parallel 2\ \text{k} \parallel 558 =$ **433 Ω**. [1]

(iv) No load: $r_d \parallel R_S = 1923\ \Omega$, so $A_v = \dfrac{1.791 \times 1.923}{1 + 1.791 \times 1.923} =$ **0.775**.

With $R_L = 1$ kΩ: $r_d \parallel R_S \parallel R_L = 657.9\ \Omega$, so $A_{v_L} = \dfrac{1.791 \times 0.6579}{1 + 1.791 \times 0.6579} =$ **0.541**. [1]

(v) $v_o = 0.541 \times 0.5 =$ **0.270 V peak**, in phase with v_i. $v_{gs} = v_i - v_o =$ **0.230 V peak**. [1]

(b)(i) No gate current flows, so R_F carries no DC current and $V_{GS} = V_{DS}$. Then

$$V_{GS} = 12 - 2\ \text{k} \times 0.5\ \text{mA/V}^2\,(V_{GS} - 2)^2$$

so $(V_{GS} - 2)^2 + V_{GS} - 12 = 0$. The roots are $V_{GS} = 4.70$ V and -1.70 V. Reject −1.70 V because it is below $V_{GS(Th)}$. [1]

$V_{GSQ} = V_{DSQ} =$ **4.70 V** and $I_{DQ} = 0.5(2.70)^2 =$ **3.65 mA**.

Saturation needs $V_{DS} \geq V_{GS} - V_{GS(Th)}$. Here $V_{DS} = V_{GS}$, which is always greater than $V_{GS} - 2$, so saturation is automatic. [1]

(ii) $g_m = 2k(V_{GSQ} - V_{GS(Th)}) = 2(0.5)(2.70) =$ **2.70 mS**. [1]

(iii) $r_d \parallel R_D = 50\ \text{k} \parallel 2\ \text{k} = 1923\ \Omega$.

$$Z_i = \frac{R_F + r_d \parallel R_D}{1 + g_m(r_d \parallel R_D)} = \frac{10\ \text{M} + 1.92\ \text{k}}{1 + 2.70\ \text{mS} \times 1923} = \mathbf{1.61\ M\Omega} \quad [1]$$

$$Z_o = R_F \parallel r_d \parallel R_D = \mathbf{1.92\ k\Omega} \quad [1]$$

$$A_v = -g_m(R_F \parallel r_d \parallel R_D) = -2.70\ \text{mS} \times 1923 = \mathbf{-5.19} \quad [1]$$

An exact nodal solution gives the same values. The approximation $Z_i \approx R_F/(1 + g_m R_D) = 1.56$ MΩ is 3 % low, because it ignores r_d. Give the mark if the

formula is shown.

> **COMMON MISTAKES:** in (a), using $Z_o = R_S$ (the follower's Z_o is dominated by $1/g_m$); putting R_L in parallel with R_S for the DC bias. In (b), writing $Z_i = R_F = 10$ MΩ (the Miller effect divides R_F by roughly $1 + |A_v|$); taking the negative root.

REREAD: E11 (source follower), E12 (drain-feedback configuration).

Question 10 (7 marks)

1. Self-bias: $V_{GS} = -1000 I_D$. The quadratic gives roots of 9.37 mA (which gives $V_{GS} = -9.37$ V, beyond V_P, so reject it) and

$I_{DQ} =$ **2.16 mA** and $V_{GSQ} =$ **−2.16 V**. [1]

$V_{DSQ} = 18 - 2.161(3.3 + 1.0) =$ **8.71 V**, so both stages are saturated. [1]

2. $g_{m0} = 2(8)/4.5 = 3.556$ mS. $g_m = 3.556\left(1 - \dfrac{2.161}{4.5}\right) =$ **1.85 mS**. [1]

3. Stage 1 is loaded by $Z_{i2} = R_G = 2.2$ MΩ:

$$A_{v1} = -g_m(r_d \parallel R_D \parallel R_G) = -1.848\ \text{mS} \times 3044\ \Omega = \mathbf{-5.63} \quad [1]$$

Stage 2 is loaded by R_L:

$$A_{v2} = -g_m(r_d \parallel R_D \parallel R_L) = -1.848\ \text{mS} \times 2336\ \Omega = \mathbf{-4.32} \quad [1]$$

4. $A_{vT} = (-5.63)(-4.32) =$ **+24.3**. The input divider is $2.2\ \text{M}/(2.2\ \text{M} + 10\ \text{k}) = 0.9955$, so $v_o/v_{sig} =$ **24.2** and $V_o = 24.2 \times 20\ \text{mV} =$ **0.484 V rms**, in phase with the source. [1]

$Z_i = R_G =$ **2.2 MΩ** and $Z_o = r_d \parallel R_D =$ **3.05 kΩ**. [1]

Without r_d, you would get $A_{v1} = -6.09$ and $A_{v2} = -4.59$. Since r_d is given, it must be used.

> **COMMON MISTAKES:** multiplying the unloaded gains (−5.63 × −5.63 = 31.7); forgetting that two inverting stages give a positive overall gain; leaving out r_d when it is given.

REREAD: E12 (cascade), E8 (loading method).

Question 11 (7 marks)

(a)(i) $R'_L = V_{CC}/I_{CQ} = 12/0.15 = 80\ \Omega$. The secondary load is $8 \parallel 8 = 4\ \Omega$. [1]

$$a = \sqrt{\frac{R'_L}{R_L}} = \sqrt{\frac{80}{4}} = \mathbf{4.47}, \quad N_1 : N_2 \approx \mathbf{4.47 : 1} \quad [1]$$

(a)(ii) $P_i = V_{CC} I_{CQ} = 1.80$ W. The maximum ideal $P_o = \frac{1}{2} V_{CC} I_{CQ} =$ **0.90 W**, so $\eta =$ **50 %**. [1]

(b)(i) $V_{L(p)}$ can reach V_{CC}:

$$P_{o(max)} = \frac{V_{CC}^2}{2R_L} = \frac{400}{16} = \mathbf{25.0\ W}, \quad P_i = \frac{2V_{CC}^2}{\pi R_L} = 31.8\ \text{W}, \quad \eta = \mathbf{78.5\%} \quad [2]$$

(At this point $P_{2Q} = 6.83$ W.)

(b)(ii) Dissipation is largest at $V_{L(p)} = 2V_{CC}/\pi =$ **12.7 V**, not at full output:

$$P_{2Q(max)} = \frac{2V_{CC}^2}{\pi^2 R_L} = \frac{800}{\pi^2 \times 8} = \mathbf{10.1\ W}\ \text{total}, \quad \mathbf{5.07\ W}\ \text{per transistor} \quad [1]$$

At that point $P_o = 10.1$ W and $P_i = 20.3$ W, so $\eta =$ **50 %**. (A numerical sweep of $P_{2Q}(V_{L(p)})$ confirms the maximum at 12.73 V.) [1]

> **COMMON MISTAKES:** using $a = R'_L/R_L$ instead of its square root; using a single speaker (8 Ω) instead of the parallel pair (4 Ω); reporting the dissipation at maximum output (6.83 W) as the maximum dissipation; giving the total dissipation when the question asks per transistor.

REREAD: E13 (power amplifiers).

After marking

Q	Topic	Your mark	Max	If under 60 %, reread
1	MCQ		8	Pack named in the table under Q1
2	Semiconductors, diode r_d and T		7	E1, E2
3	Clipper and clamper		5	E3
4	Bridge and C filter		5	E3
5	pnp bias, β, saturation		8	E4, E5
6	Collector feedback, CB		8	E6, E7
7	Current mirror		5	E8
8	D-MOSFET		8	E10, E11
9	Source follower, drain feedback		12	E11, E12
10	JFET cascade		7	E12, E8
11	Power amps		7	E13
	Total		**80**	

Diodes and BJT total (Q2–Q6 plus Q1 I–III) out of 36: ____. FETs, multistage, op-amp and power total (Q7–Q11 plus Q1 IV–VIII) out of 44: ____. Spend tonight's remaining time on the weaker half, using the packs above. Do not start new material.

Night-before checklist

Pack the bag

- ☐ Calculator (with its model written down for the cover sheet) and spare batteries
- ☐ Printed open-book reference sheet, with the formula tables from this mock and from REFE
- ☐ Lecture notes for Weeks 6–8, WS7–10, and your Test 1 reference (Weeks 1–5), all tabbed
- ☐ Pens, pencil, ruler, eraser (you will be sketching waveforms and load lines)

Ten reflexes to have ready (read once, then sleep)

1. **Units first.** Convert °C to K. Put semiconductor lengths in cm. Use k in eV/K inside $e^{-E_g/2kT}$.
2. **Diodes:** write the state you assume, find where it changes (v_i at the transition), then sketch v_o under v_i and label the levels and angles in words.
3. **Full-wave means** $2f$, and a bridge loses two diode drops while its PIV is only one secondary peak.
4. **BJT DC:** use Thevenin and the exact I_B formula. For a pnp, flip every arrow. Always check V_{CE} (or V_{EC}) against 0.2 V. If it fails, redo the analysis with the saturation model; do not keep βI_B.
5. **Small-signal:** do DC first, then r_e or g_m, then Z_i, Z_o, A_v, then the load, then the source divider. Watch the sign: CE, CS and drain feedback invert; CB and followers do not.
6. **FET quadratics:** keep the root with V_{GS} between V_P (or $V_{GS(Th)}$) and the right side of the curve. A D-MOSFET may have a positive V_{GS} and $I_D > I_{DSS}$, and that is fine.
7. $g_m = g_{m0}(1 - V_{GSQ}/V_P)$. Use $r_d = 1/y_{os}$ whenever it is given.
8. **Cascades:** load each stage with the next stage's Z_i, then multiply. Two inverting stages give a positive gain.
9. **Power:** $R'_L = a^2 R_L$. Class A transformer-coupled η_{max} is 50 %; class B is 78.5 %. Class B dissipation peaks at $V_{L(p)} = 2V_{CC}/\pi$.
10. **Digital and op-amp:** $NM_L = V_{IL} - V_{OL}$, $NM_H = V_{OH} - V_{IH}$, and CMRR(dB) $= 20\log(A_d/A_c)$.

In the exam room

- ☐ Read the whole paper in the first 5 minutes. Mark the questions you can bank quickly.
- ☐ Do the MCQs and the short calculations first, then the long FET and BJT questions.
- ☐ Write the formula, then the substitution, then the answer with units, for every part. Method marks carry you when a number slips.
- ☐ If a question needs a figure value you cannot read, state your assumption in one line and carry on.
- ☐ Leave 5 minutes at the end to check signs, units and root choices.

Sleep by a sensible hour. A rested brain is worth more than one more pack tonight.